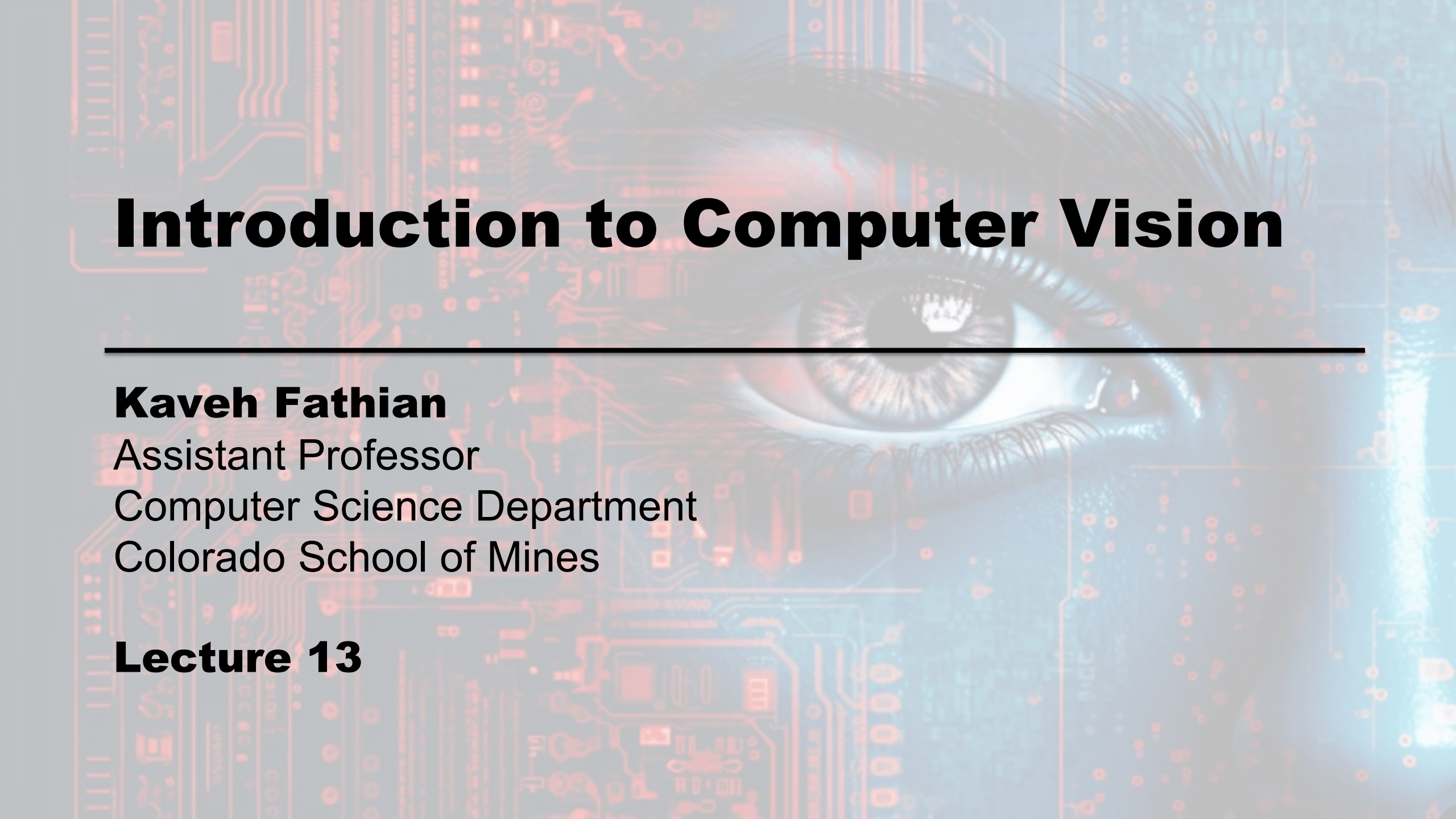


The background of the slide is a composite image. It features a close-up of a human eye, looking directly forward. Overlaid on this image is a complex, glowing blue and white digital circuit pattern, resembling a printed circuit board (PCB) or a network diagram. The eye's iris is a light blue color, and the eyelashes are dark and well-defined. The overall aesthetic is high-tech and futuristic.

Introduction to Computer Vision

Kaveh Fathian

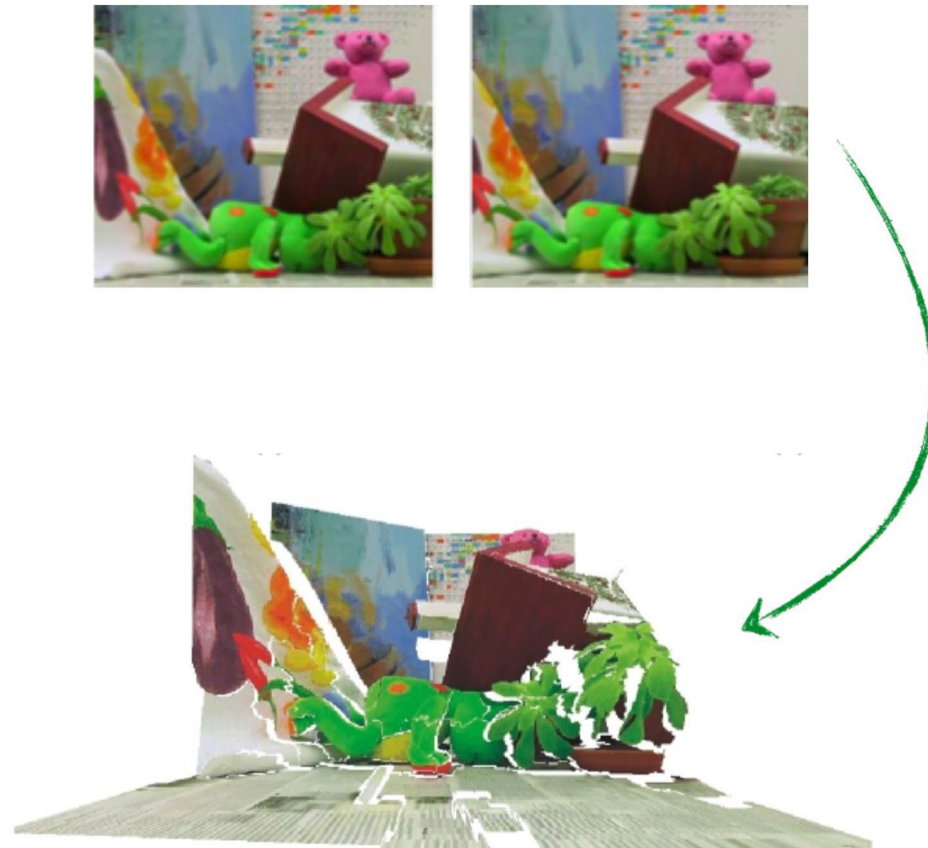
Assistant Professor

Computer Science Department

Colorado School of Mines

Lecture 13

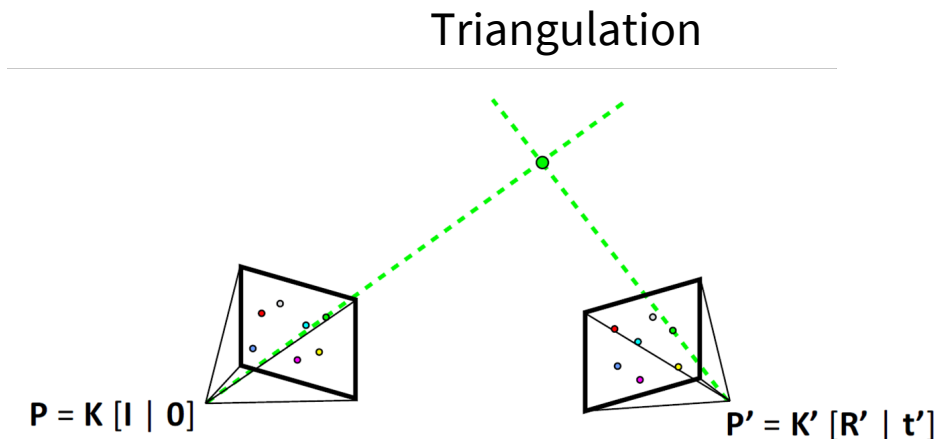
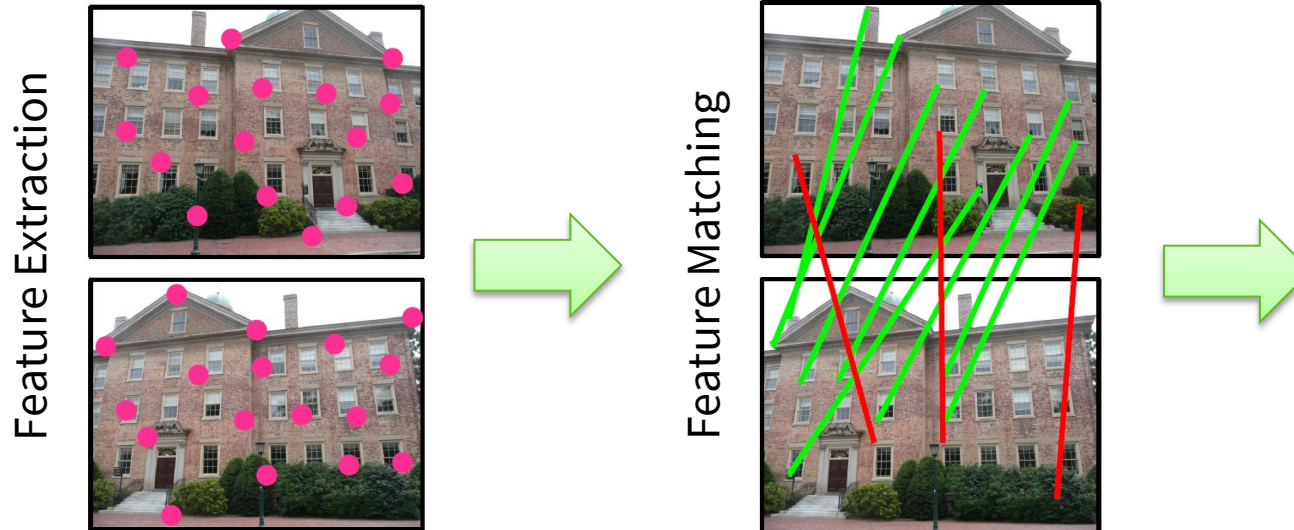
Stereo Vision



Stereo Vision

Can we reconstruct a 3D depth map from 2 images?

- Use **2-view geometry**:
 1. Extract features
 2. Match features
 3. Find essential/homography matrix (via RANSAC)
 4. Recover camera matrix (R, t)
 5. Triangulate 3D points



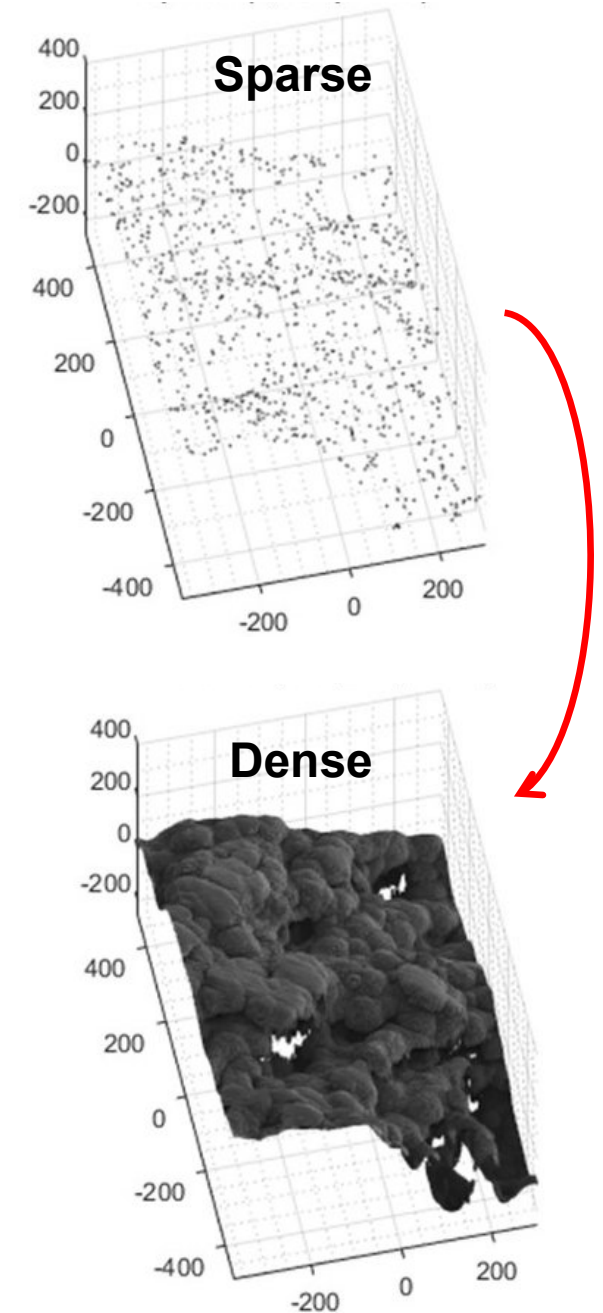
Stereo Vision

Issues of 2-view geometry:

- Sparse 3D map
- Time consuming to find (correct) corresponding features
- Time consuming to triangulate

Stereo vision: Given camera matrix (calibration, R, t), stereo reconstruction solves issues:

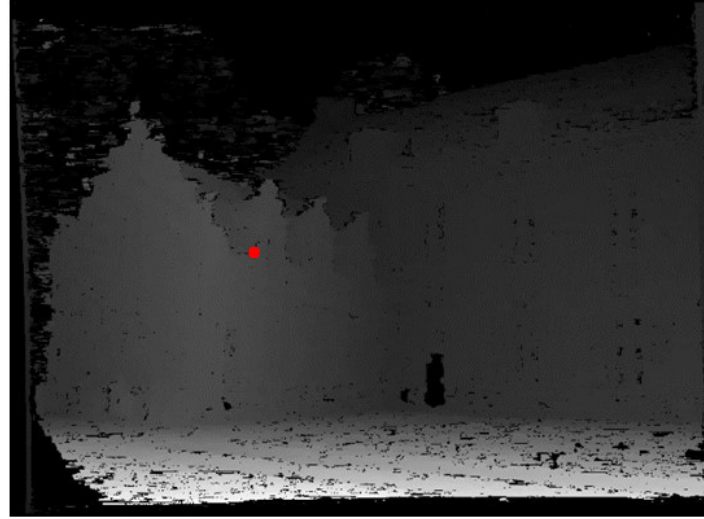
- Dense 3D map
- Fast computation



Depth from Disparity



image $I(x,y)$



Disparity map $D(x,y)$

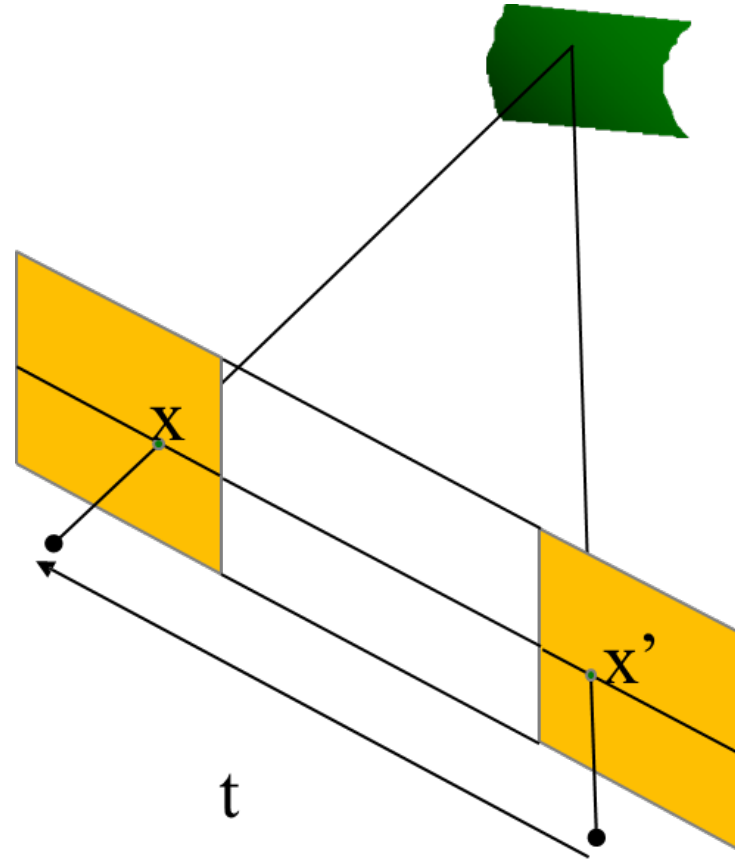


image $I'(x',y')$

$$(x', y') = (x+d, y)$$

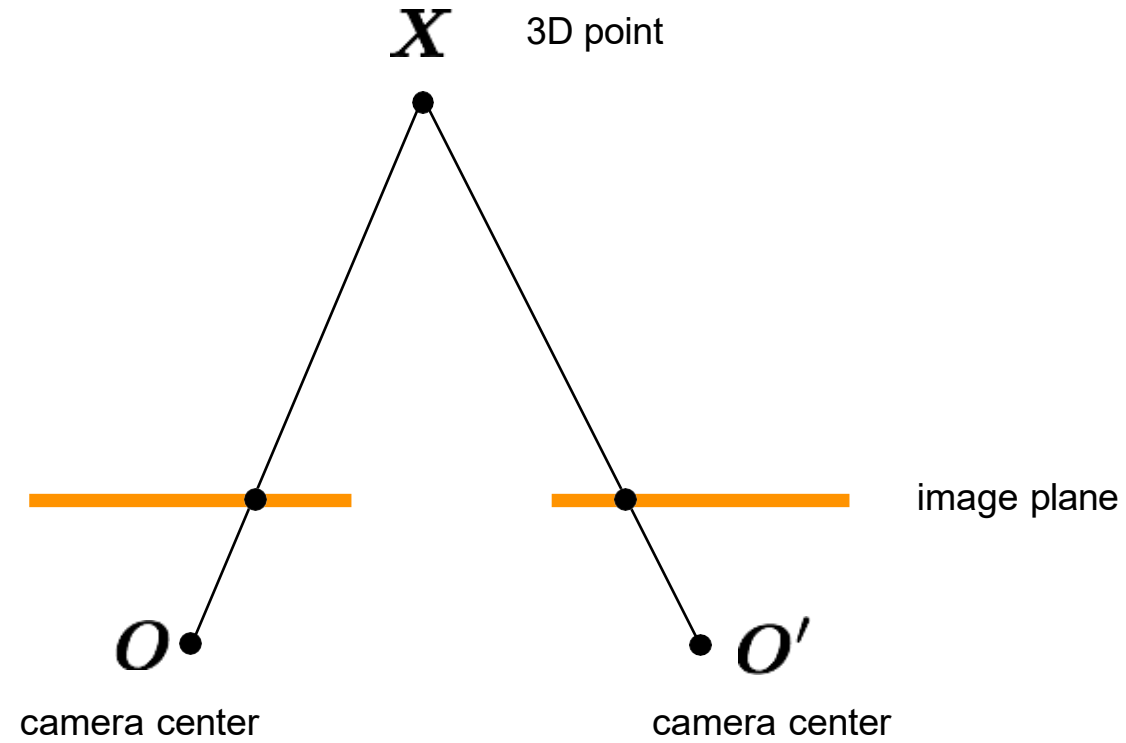
If we could find the **corresponding points** in two images, we could **estimate relative depth**...

Depth from Disparity

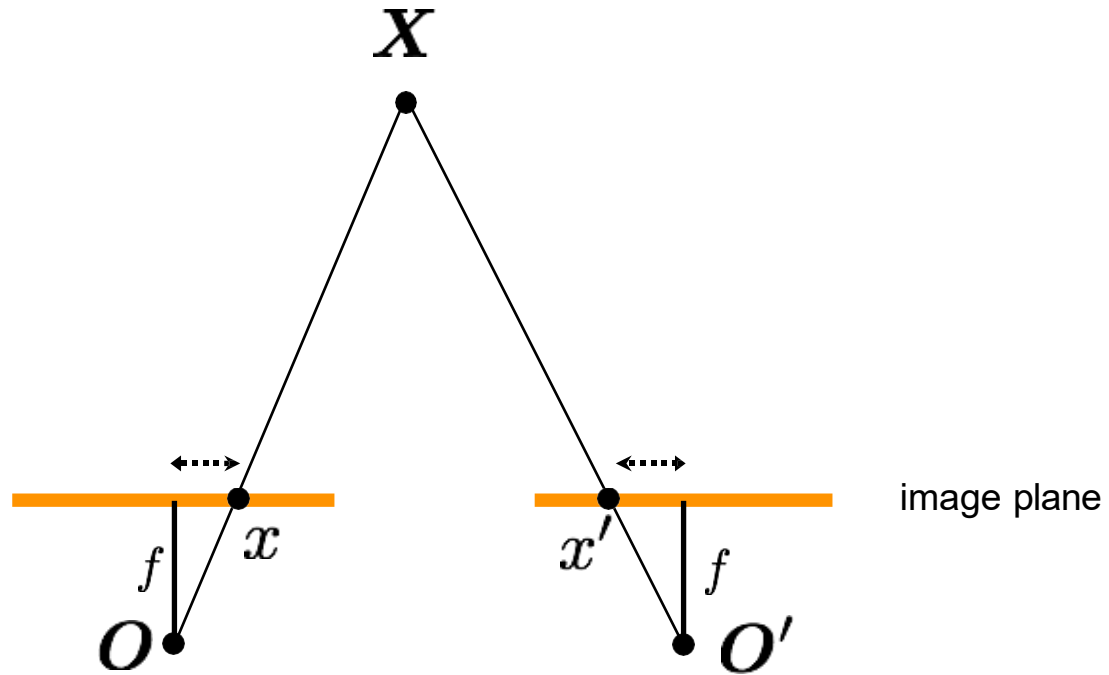


- Assume 2 cameras with CCDs on the **same plane**
- Assume **known correspondence** between images of a 3D point

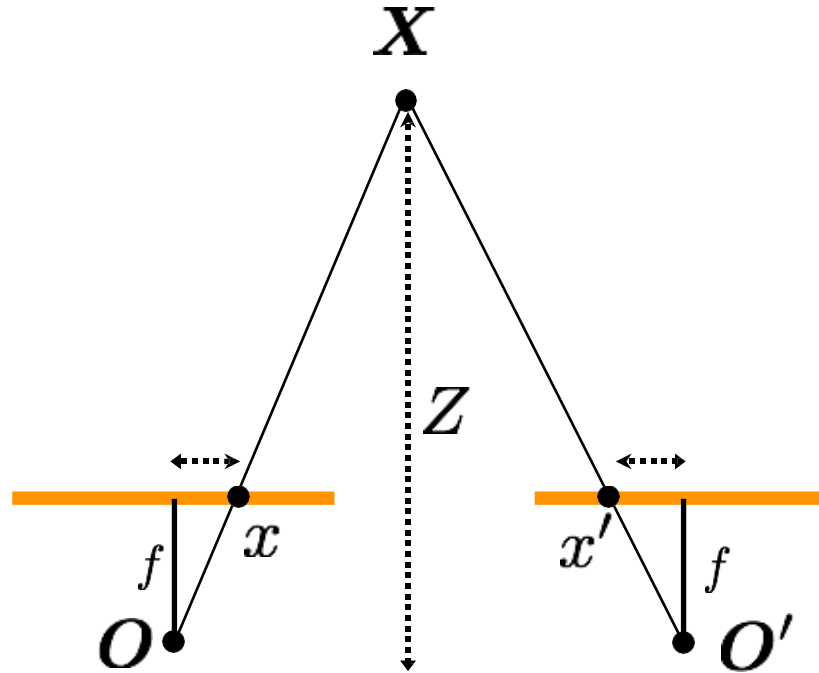
Depth from Disparity



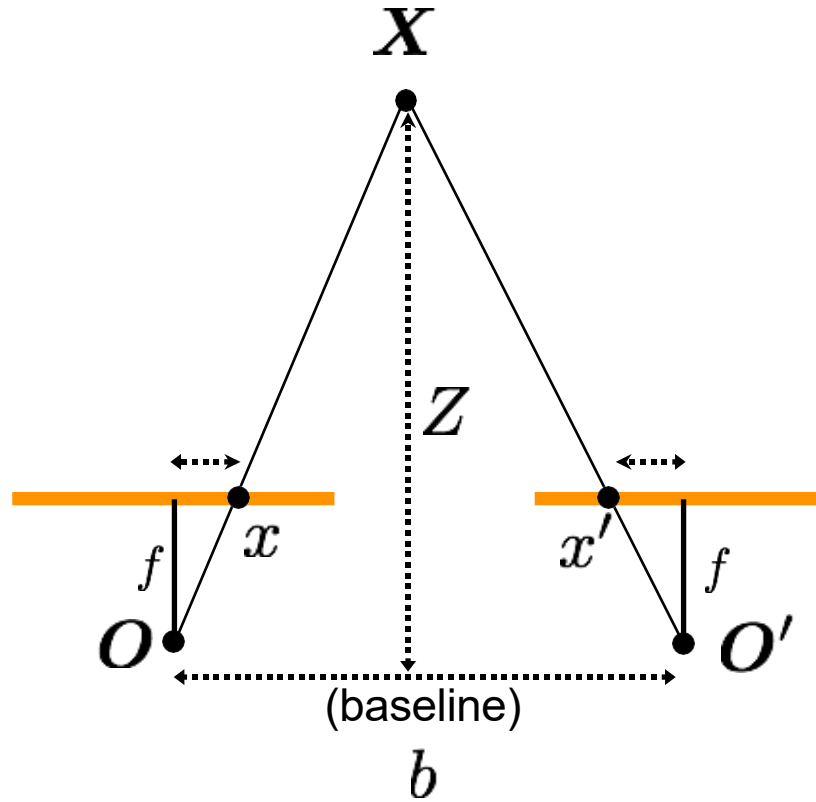
Depth from Disparity



Depth from Disparity

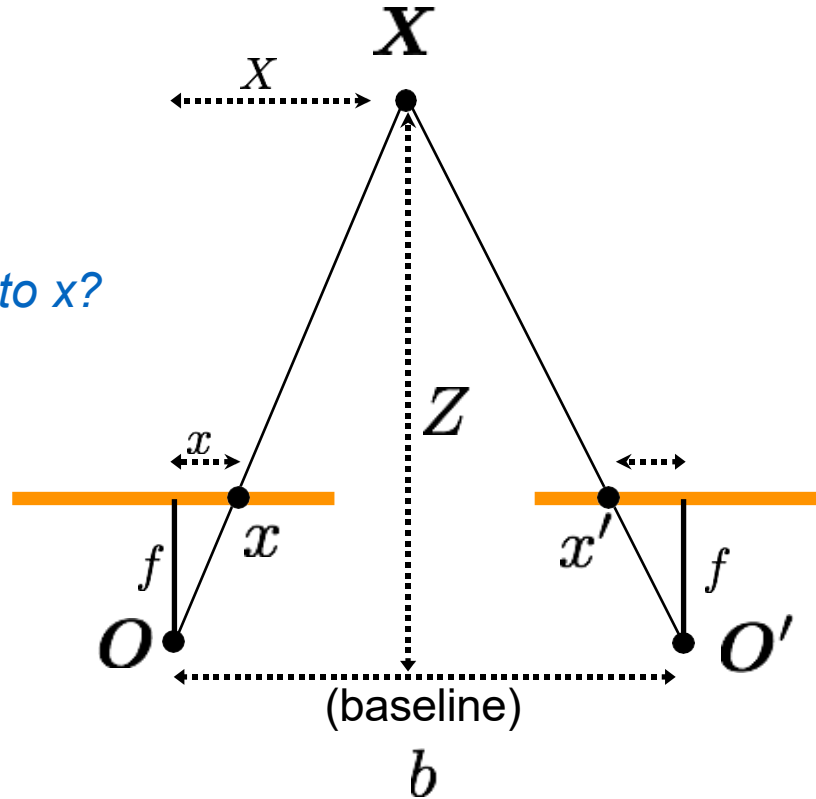


Depth from Disparity



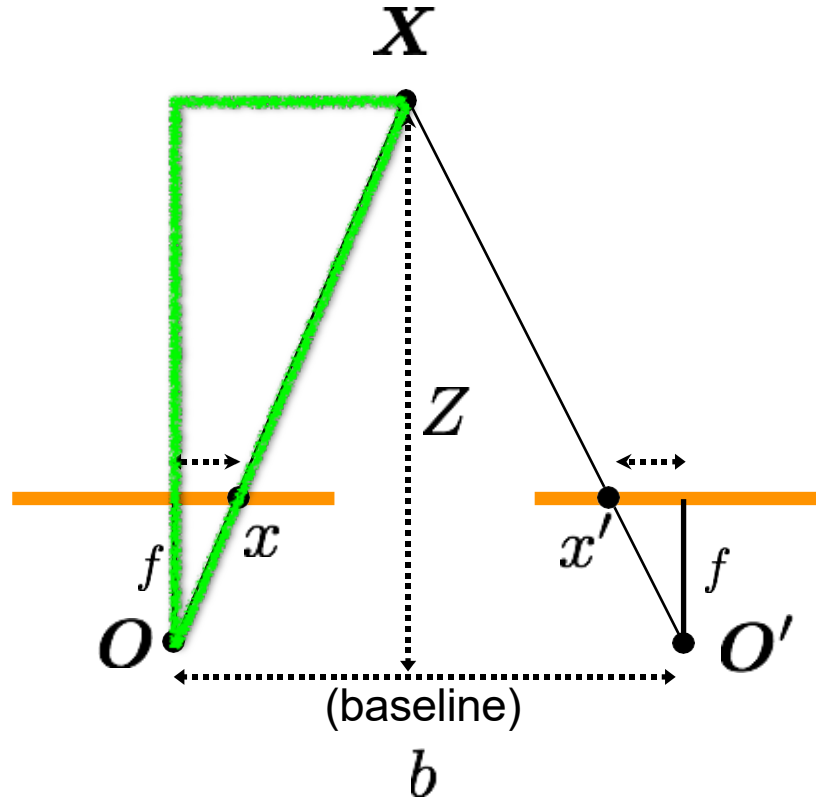
Depth from Disparity

How is X related to x ?



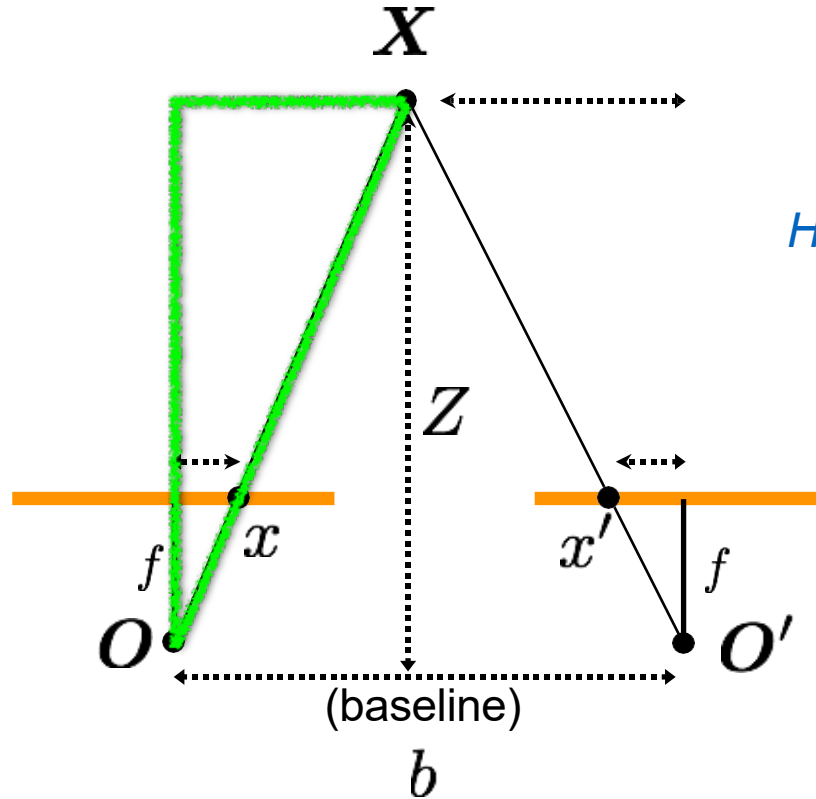
Depth from Disparity

$$\frac{X}{Z} = \frac{x}{f}$$

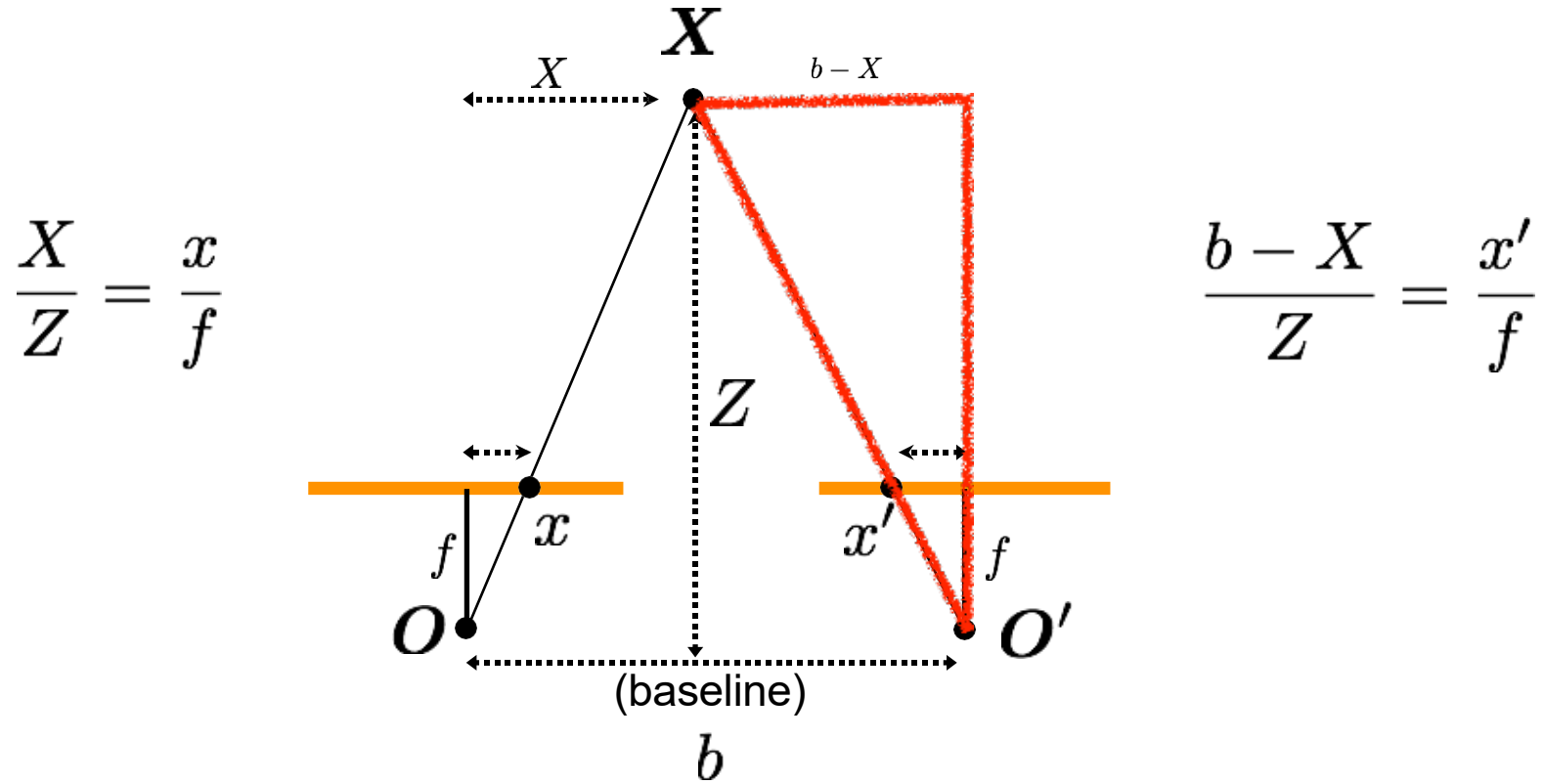


Depth from Disparity

$$\frac{X}{Z} = \frac{x}{f}$$

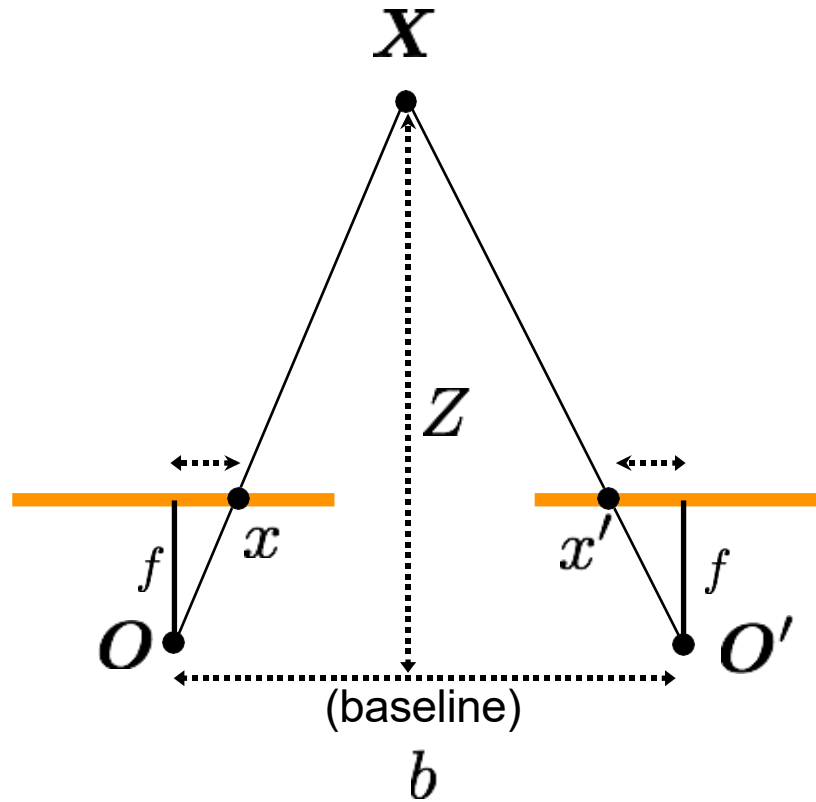


Depth from Disparity



Depth from Disparity

$$\frac{X}{Z} = \frac{x}{f}$$



$$\frac{b - X}{Z} = \frac{x'}{f}$$

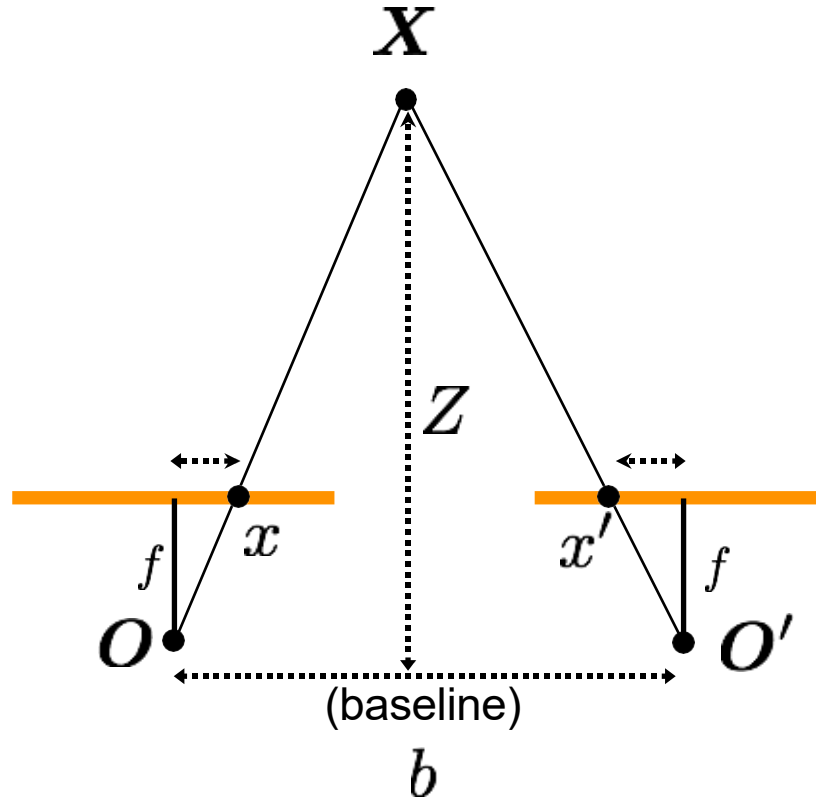
Disparity

$$d = x - x' \quad (\text{wrt to camera origin of image plane})$$

$$= \frac{bf}{Z}$$

Depth from Disparity

$$\frac{X}{Z} = \frac{x}{f}$$



$$\frac{b - X}{Z} = \frac{x'}{f}$$

Disparity

$$d = x - x'$$

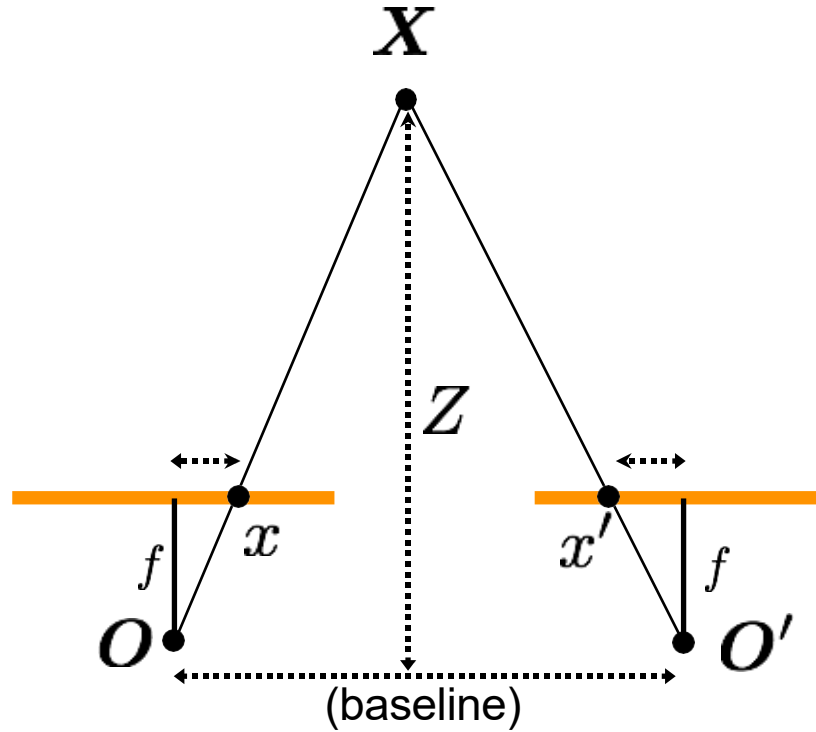
$$= \frac{bf}{Z}$$

inversely proportional
to depth

Depth from Disparity

$$\frac{X}{Z} = \frac{x}{f}$$

$$\frac{b - X}{Z} = \frac{x'}{f}$$



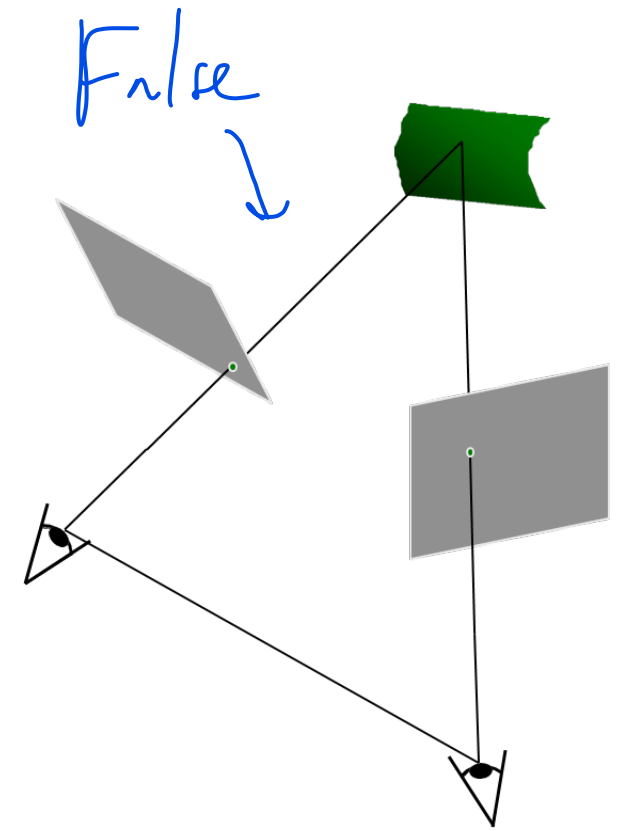
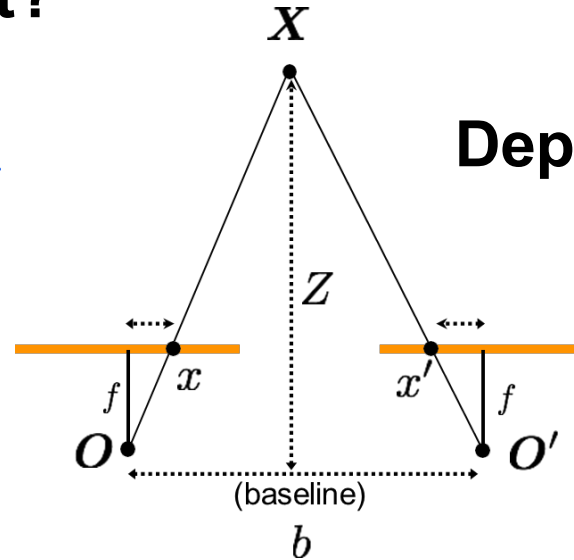
b
 (baseline)
Depth
 $Z = \frac{bf}{d}$
 $\rightarrow$ focal length (known)
 $\rightarrow$ disparity (known)

Handwritten annotations: "known" with an arrow pointing to b , and "unknown" with an arrow pointing to Z .

Stereo Vision

- We assume 2 cameras with CCDs on the **same plane**
 - **What if they are not?**

only for parallel

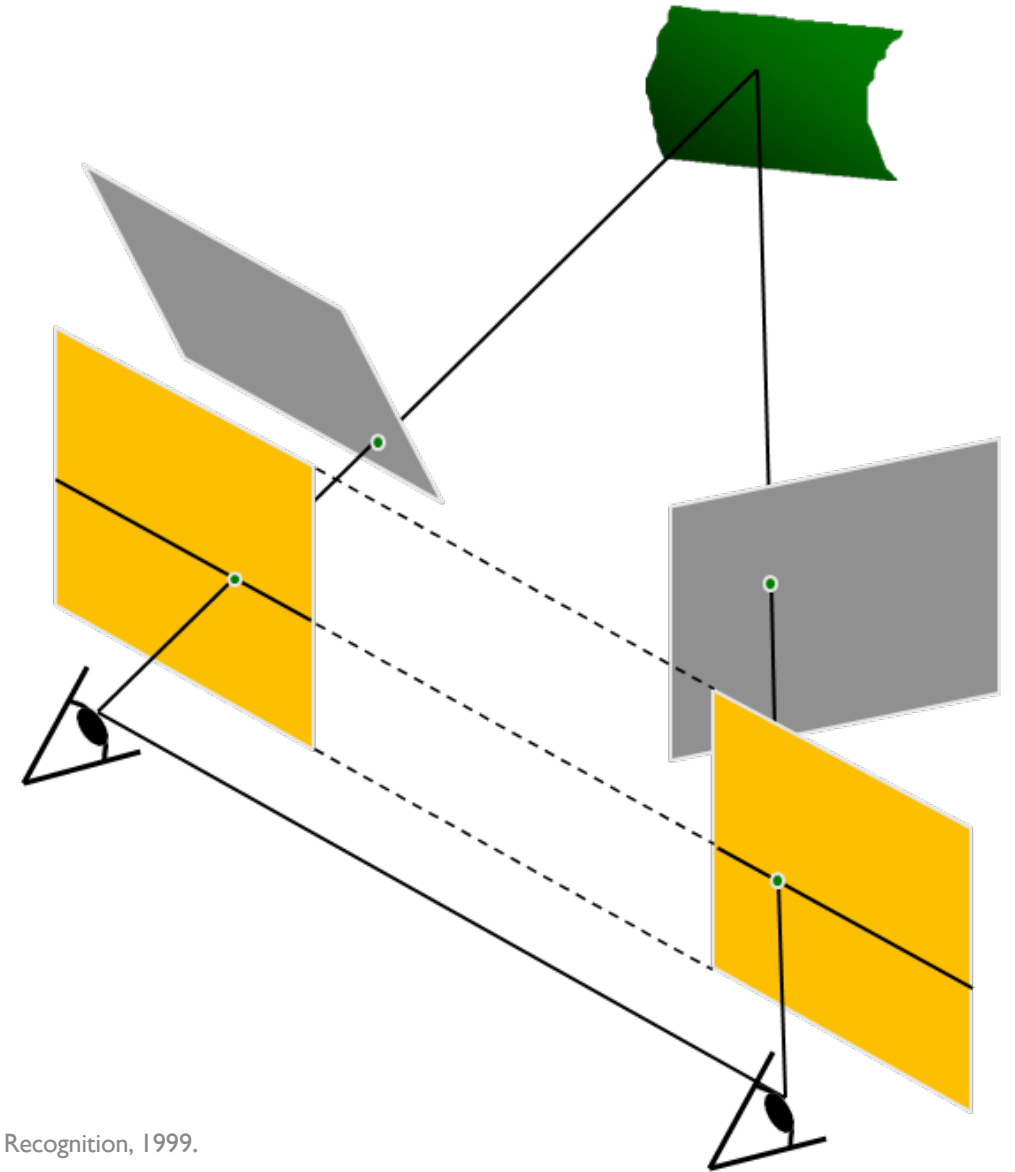


- We assumed **known correspondence** between images of 3D points
 - **How to find (correct) correspondences?**



Stereo Image Rectification

- Reproject image planes onto a common plane parallel to the line between camera centers
- Pixel motion is horizontal after this transformation
- Two homographies (3x3 transform), one for each input image reprojection

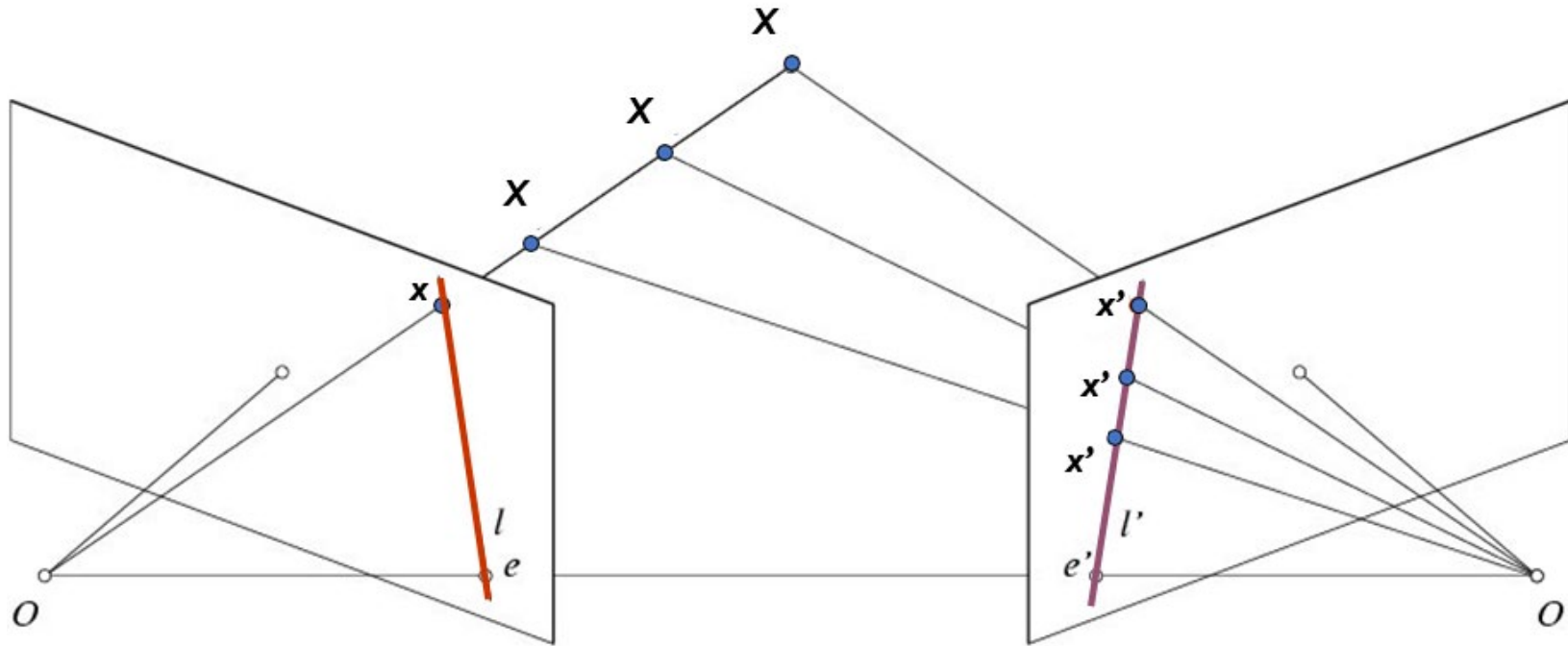


C. Loop and Z. Zhang. [Computing Rectifying Homographies for Stereo Vision](#). IEEE Conf. Computer Vision and Pattern Recognition, 1999.

Finding Correspondences using Epipolar Constraint

How to **efficiently** find correspondences?

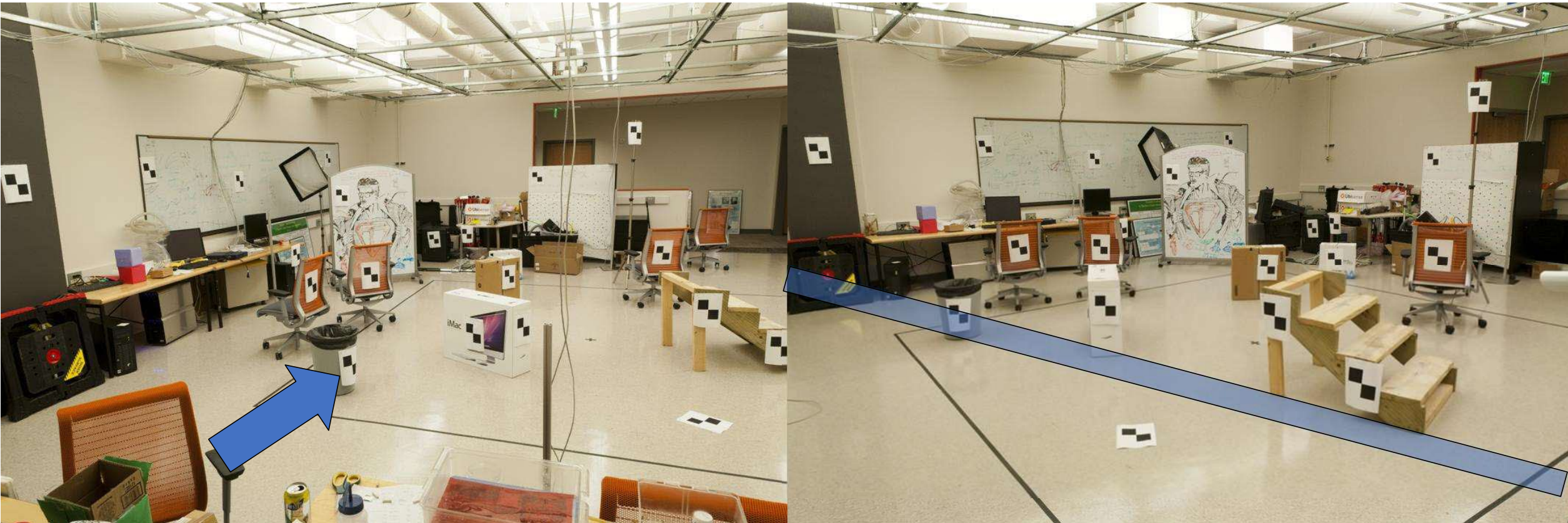
- **Key Idea:** Epipolar Constraint



Potential matches for x' have to lie on the corresponding line l .

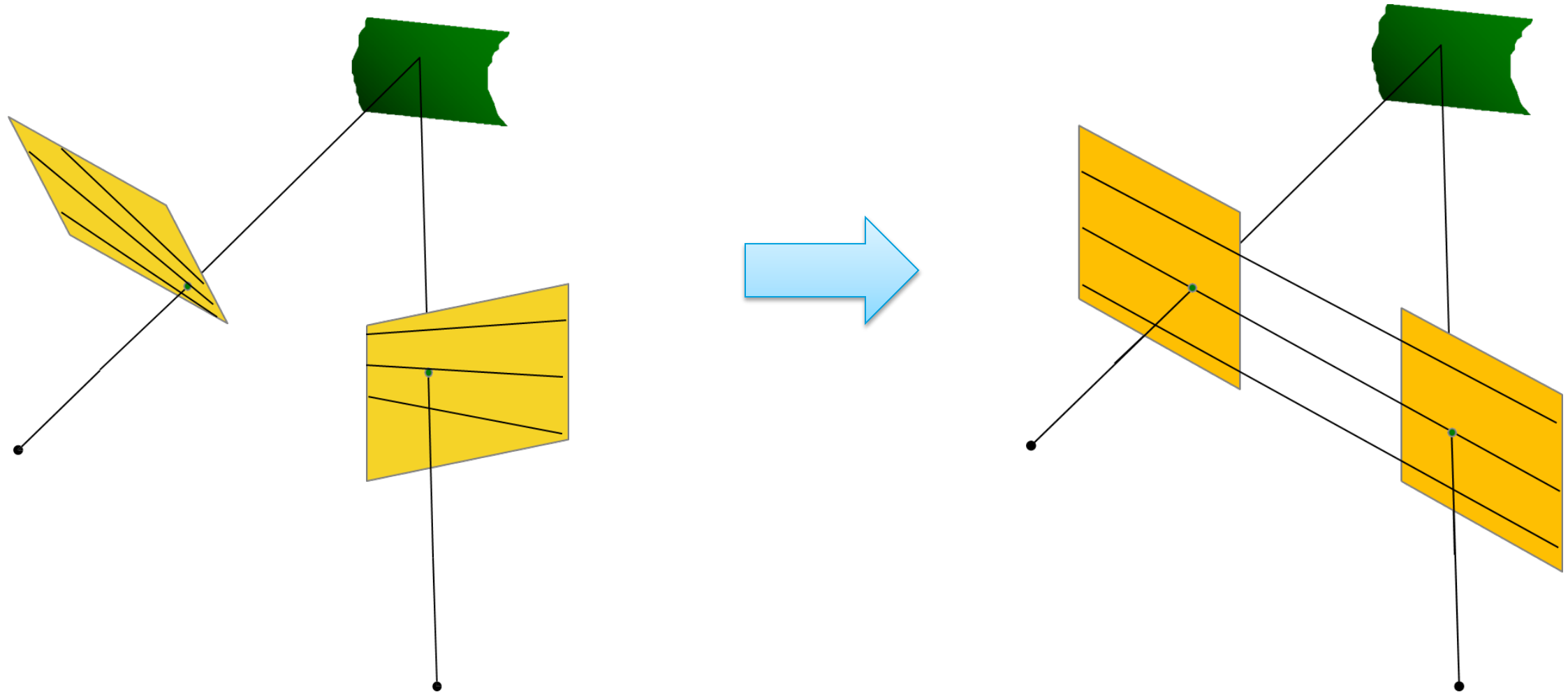
Potential matches for x have to lie on the corresponding line l' .

Finding Correspondences using Epipolar Constraint

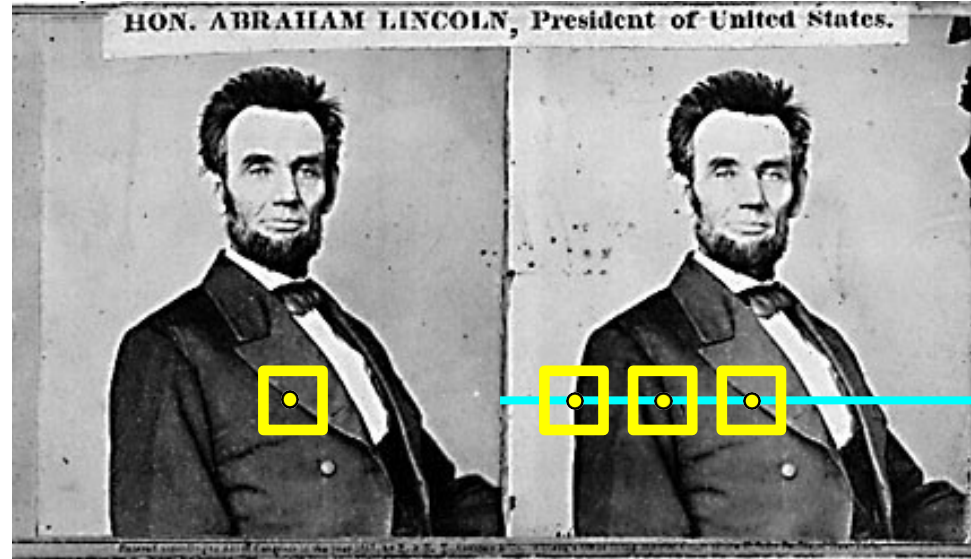


Finding Correspondences using Epipolar Constraint

After stereo rectification, epipolar lines become horizontal:



Stereo Vision



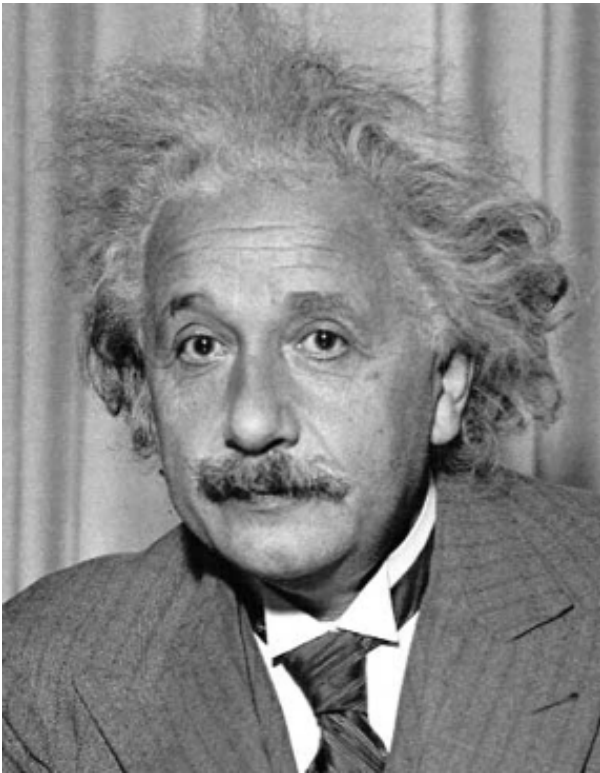
1. Rectify images
(make epipolar lines horizontal)
2. For each pixel (or window)
 - a. Find epipolar line
 - b. Scan line for best match
 - c. Compute depth from disparity

How would
you do this?

$$Z = \frac{bf}{d}$$

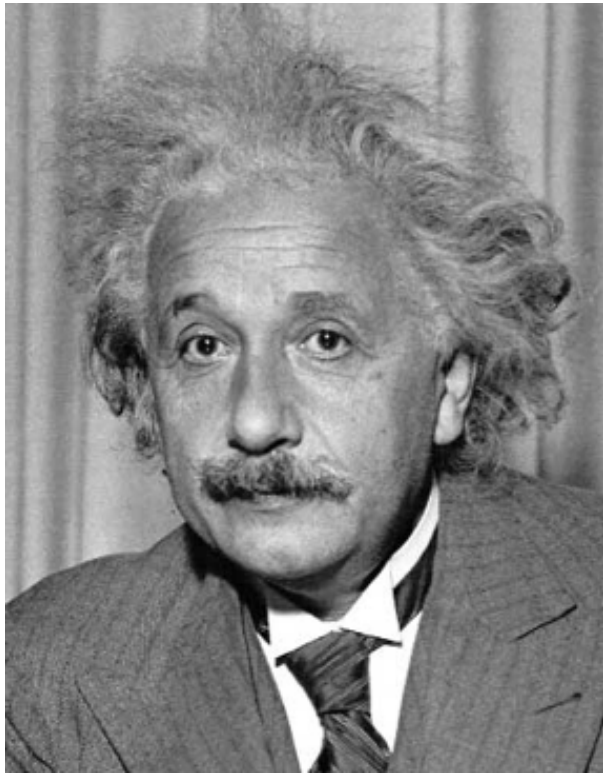
Template Matching

How do we detect the template  in the following image?



Template Matching

How do we detect the template  in the following image?

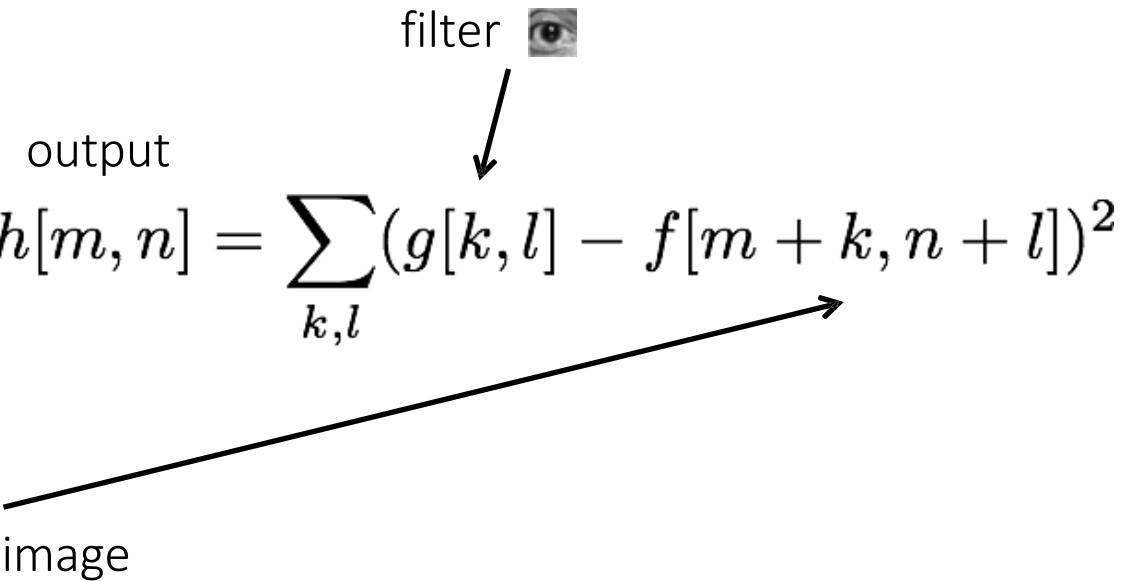


output

filter 

$$h[m, n] = \sum_{k, l} (g[k, l] - f[m + k, n + l])^2$$

image

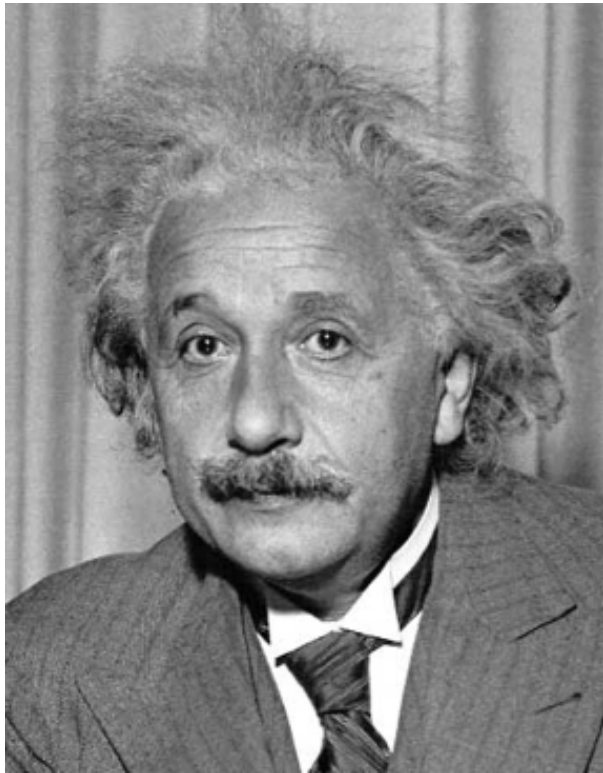


What will the output look like?

Solution 3: Use sum of squared differences (SSD).

Template Matching

How do we detect the template  in the following image?



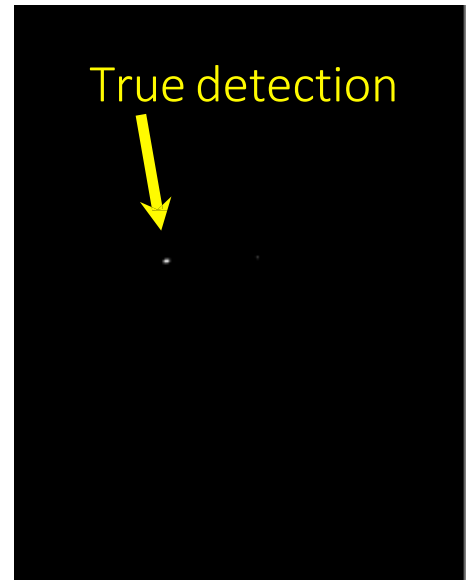
output

filter 

$$h[m, n] = \sum_{k, l} (g[k, l] - f[m + k, n + l])^2$$

image

The diagram shows an arrow labeled 'image' pointing from the Einstein portrait to the equation, and another arrow labeled 'filter' pointing from the eye icon to the equation.

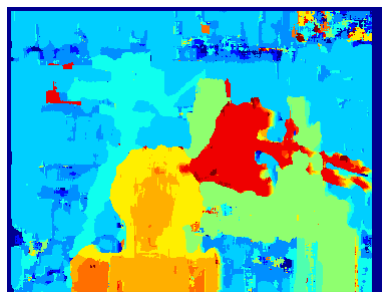


Solution 3: Use sum of squared differences (SSD).

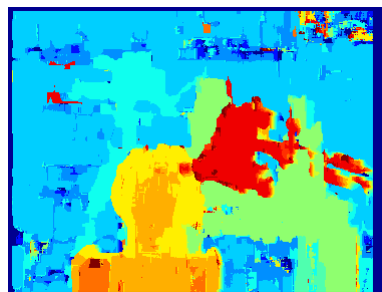
What could go wrong?

Similarity Metrics

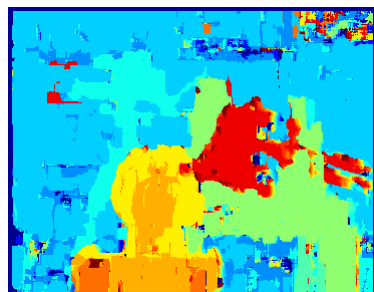
Similarity Measure	Formula
Sum of Absolute Differences (SAD)	$\sum_{(i,j) \in W} I_1(i,j) - I_2(x+i, y+j) $
Sum of Squared Differences (SSD)	$\sum_{(i,j) \in W} (I_1(i,j) - I_2(x+i, y+j))^2$
Zero-mean SAD	$\sum_{(i,j) \in W} I_1(i,j) - \bar{I}_1(i,j) - I_2(x+i, y+j) + \bar{I}_2(x+i, y+j) $
Locally scaled SAD	$\sum_{(i,j) \in W} I_1(i,j) - \frac{\bar{I}_1(i,j)}{\bar{I}_2(x+i, y+j)} I_2(x+i, y+j) $
Normalized Cross Correlation (NCC)	$\frac{\sum_{(i,j) \in W} I_1(i,j) \cdot I_2(x+i, y+j)}{\sqrt{\sum_{(i,j) \in W} I_1^2(i,j) \cdot \sum_{(i,j) \in W} I_2^2(x+i, y+j)}}$



SAD



SSD

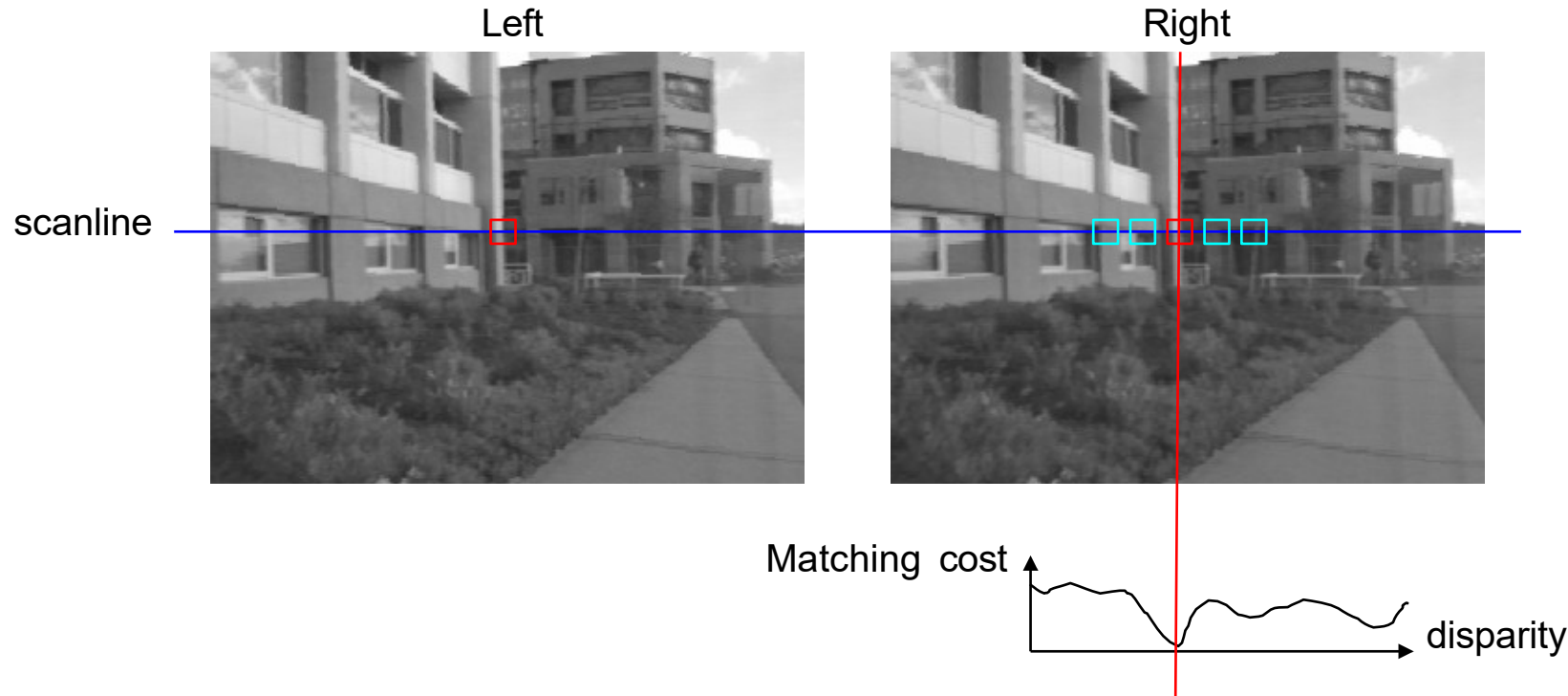


NCC



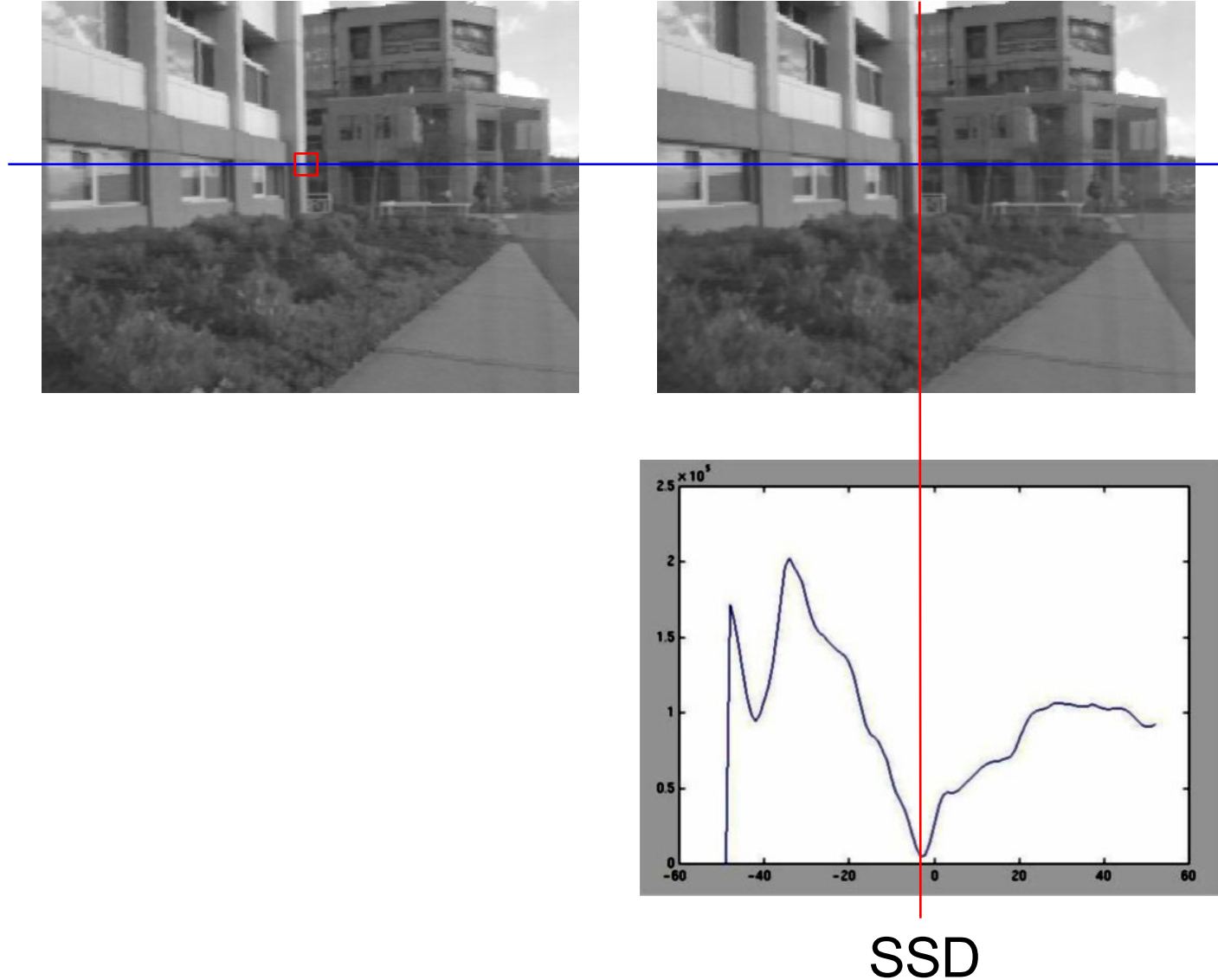
Ground truth

Stereo Block Matching

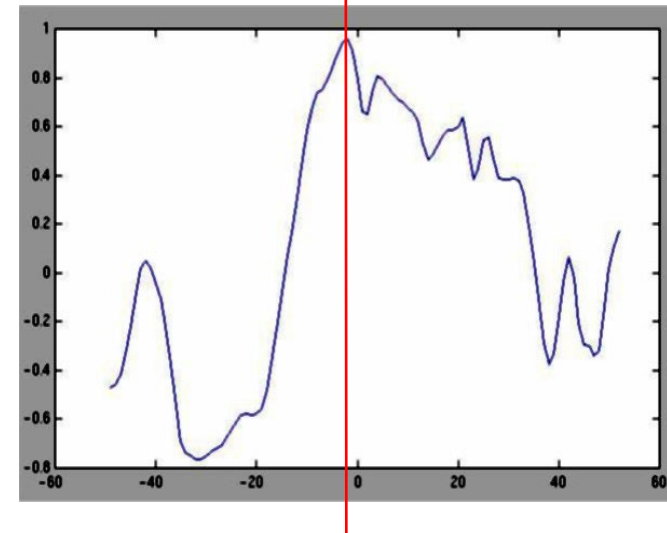


- Slide a window along the epipolar line and compare contents of that window with the reference window in the left image
- Matching cost: SSD or normalized correlation

Sum of Squared Differences



Normalized cross-correlation



Effect of window size



$W = 3$



$W = 20$

Effect of window size



$W = 3$

Smaller window

- + More detail
- More noise

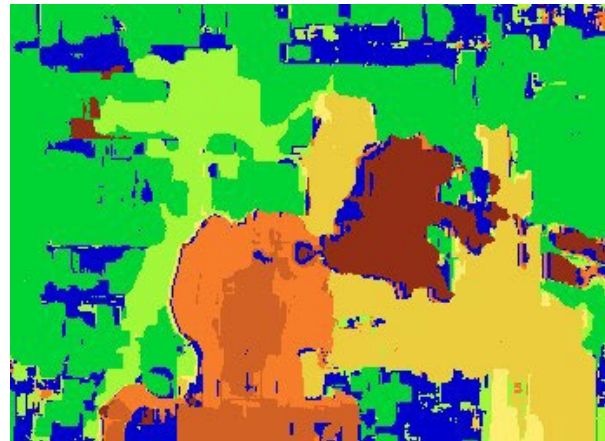
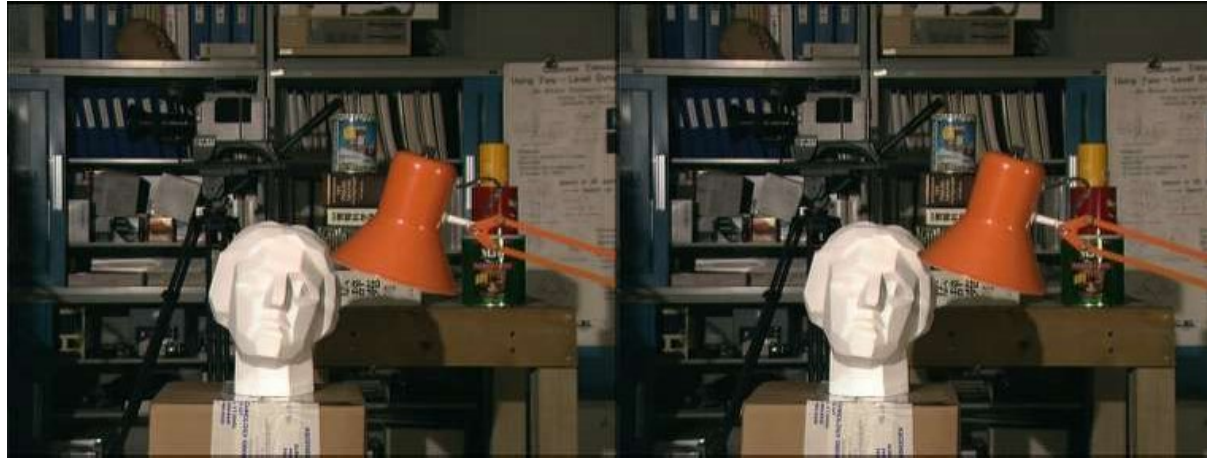


$W = 20$

Larger window

- + Smoother disparity maps
- Less detail
- Fails near boundaries

Tsukuba Test Scene



Window-based matching
(best window size)



'Ground truth'

For the latest and greatest: <https://vision.middlebury.edu/stereo/eval3/>